

CS3300 - GCP Deployment from Cloud Shell

Complementary Student Notes | Clone GitHub Repository -> App Engine

Use this workflow when you do not want to install the Google Cloud CLI on your local computer. Cloud Shell is a browser-based terminal provided by Google and already includes gcloud.

1. Open Cloud Shell

Open <https://console.cloud.google.com/> and select your CS3300 GCP project. Click the Activate Cloud Shell icon (>) in the top-right corner. A terminal opens at the bottom of the browser.

Key idea: Cloud Shell is a separate environment from your laptop. Files on your laptop are not automatically available in Cloud Shell.

2. Verify the GCP Project

```
gcloud config get-value project
```

If needed, connect to the correct project:

```
gcloud init
```

Key idea: Always verify the project before deploying.

3. Clone the Team Repository

```
git clone <repository-url>  
cd <repository-folder>
```

The repository should contain pom.xml and the App Engine deployment configuration.

4. Verify the Project Files

```
ls  
ls src/main  
cat app.yaml
```

Key idea: Cloud Shell now contains a copy of the GitHub repository. This is why deployment from Cloud Shell does not require the project files to be on your laptop.

5. Build and Test

```
mvn clean package
```

6. Deploy

```
gcloud app deploy
```

Read the displayed project, application, and region information before confirming.

7. Verify

```
gcloud app browse
```

Then append /students or your appropriate API path to the deployed URL.

8. What Changed?

Only the workflow changed:

```
GitHub -> Cloud Shell -> gcloud app deploy -> App Engine
```

The application architecture and deployment target are the same as the local-CLI workflow.

Local CLI vs. Cloud Shell

	Local computer	Cloud Shell
Where is code?	Local VS Code folder	Cloned GitHub repository
gcloud install?	Required	Already available
Build command	mvn clean package	mvn clean package
Deploy	gcloud app deploy	gcloud app deploy